

# Write-up on Runge's Theorem

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Since we will need to use Runge's theorem to embed an algebraic surface to the projective space and Runge's theorem is not a standard topic in a one-variable complex analysis class, therefore we include it in a write-up so that you can read this when you need Runge's theorem. Basically we are following [NN01]. Let's state all the versions of Runge's theorem before we begin to prove them:

**Theorem 0.1. (Runge's Theorem: First Form)** Let  $\Omega$  be open in  $\mathbb{C}$  and let  $K$  be a compact subset of  $\Omega$ , then the following statements are equivalent:

1.  $\rho(\mathcal{H}(\Omega))$  is dense in  $\mathcal{O}(K)$ .
2. No connected components of  $\Omega \setminus K$  is relatively compact in  $\Omega$ .
3. For any  $a \in \Omega$  and  $a \notin K$ , there exists a  $f \in \mathcal{H}(\Omega)$  with:

$$|f(a)| > |f|_K$$

Here  $\mathcal{H}(\Omega)$  are just holomorphic functions on  $\Omega$  and we will explain what is  $\rho$  and what is  $| \cdot |_K$  latter.

**Theorem 0.2. (Runge's Theorem: Second Form)** Let  $\Omega' \subset \Omega$  be two open sets in  $\mathbb{C}$ , then  $(\Omega, \Omega')$  is a **Runge's pair** iff no connected component of  $\Omega \setminus \Omega'$  is relatively compact.

We will define what is a Runge's pair later but simply speaking, what this is saying is that holomorphic functions in  $\Omega'$ , the smaller open set can be approximated using restrictions of holomorphic functions on a larger open set.

**Theorem 0.3. (Runge's Theorem: Classical Form)** Let  $\Omega$  be open in  $\mathbb{C}$  and let  $\mathbb{C} \setminus \Omega = \bigcup_{\alpha \in A} C_\alpha$  be the decomposition into connected components. The  $A' \subset A$  be set where  $A' = \{\alpha \in A : C_\alpha \text{ is compact.}\}$ , and for each  $\alpha \in A'$ , choose a point  $a_\alpha \in C_\alpha$ .

Then for any  $f \in \mathcal{H}(\Omega)$ , it can be approximated uniformly on any compact subset of  $\Omega$  by rational functions all of whose poles are contained in the set  $\{a_\alpha\}_{\alpha \in A'}$ .

A corollary of this theorem is:

**Corollary 0.4.** Let  $\Omega$  be open in  $\mathbb{C}$ . Then the set of restrictions to  $\Omega$  of polynomials in  $z$  is dense in  $\mathcal{H}(\Omega)$  iff  $\mathbb{C} \setminus \Omega$  has no compact connected component.

Before we begin our study on Runge's theorem, let's examine some preliminaries:

## 1 The $\frac{\partial u}{\partial \bar{z}} = \phi$ Equation

In our proof of Runge's theorem, one important tool is to understand the so called **inhomogeneous version of Cauchy's Integral formula**. To be able to set this up, we will have to examine the following partial differential equation on an open subset  $\Omega$  of  $\mathbb{C}$ :

$$\frac{\partial u}{\partial \bar{z}} = f$$

where  $f$  is a smooth function on  $\Omega$ , or we can write is as  $f \in C^\infty(\Omega)$ . Here the differential operator  $\frac{\partial}{\partial \bar{z}}$  just means:

$$\frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left( \frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right)$$

This equation is sometime called the  $\bar{\partial}$  equation and as you will see later, a lot of complex geometry actually hides behind this equation. Turns that we won't be able to solve it in the full generality without Runge's theorem, however the study of this partial differential equation helps us prove Runge's theorem, which in turn solves the equation.

We will need the following fact on smooth functions which basically comes from the existence of a partition of unity:

**Theorem 1.1.** Let  $\Omega$  be open in  $\mathbb{R}^n$  and let  $X \subset \Omega$  be a closed subset which is contained in some  $U$  open in  $\Omega$ . Then there is a  $\phi \in C^\infty(\Omega)$  with  $0 \leq \phi(x) \leq 1$  for all  $x \in \Omega$  such that  $\phi|_X = 1$  and  $\phi|_{\Omega \setminus U} = 0$ .

What this function is telling us is that we can create a bump function which takes exactly the value 1 on some closed subset.

We will also need the following theorem on some integrations:

**Lemma 1.2.** Let  $R$  be a closed rectangle in  $\mathbb{C}$  and let  $f \in C^\infty(R)$  with values in  $\mathbb{C}$ , which means that there is an open neighborhood of  $R$  where  $f$  is smooth as a function from that open neighborhood to  $\mathbb{R}^2$ . Then we have the following integral formula:

$$\iint_R \frac{\partial f}{\partial \bar{z}} dz \wedge d\bar{z} = - \int_{\partial R} f dz$$

Here since  $dz = dx + idy$  and  $d\bar{z} = dx - idy$ , we know that  $dz \wedge d\bar{z} = -2idx \wedge dy$ . Therefore you can just think of this as the Lebesgue integration with the standard planar Lebesgue measure.

*Proof.* Suppose that  $R = [a, b] \times [c, d]$ , then we know that:

$$\iint_R \frac{\partial f}{\partial \bar{z}} dz \wedge d\bar{z} = (-i) \int_c^d \left( \int_a^b \frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} dx \right) dy$$

We know that if we compute:

$$\int_c^d \int_a^b \frac{\partial f}{\partial x} dx dy$$

It's just:

$$\int_c^d f(b, y) - f(a, y) dy$$

Let call the boundary of  $R$ , positively oriented by  $\gamma_1, \dots, \gamma_4$ , in counter clockwise. Recall that if we parametrize the left and right vertical side of the rectangle by  $t \mapsto b + ti$  and  $t \mapsto a + (d + c)i - ti$ , then we know that:

$$\int_{\gamma_2} f(z) dz = i \int_c^d f(b, t) dt, \quad \int_{\gamma_4} f(z) dz = -i \int_c^d f(a, d + c - t) dt$$

Notice that for the second term, we do a change of variable by  $y = d + c - t$ , we know that  $dy = -dt$  and the integral is:

$$-i \int_c^d f(a, d + c - t) dt = -i \int_d^c f(a, y) - dy = -i \int_c^d f(a, y) dy$$

Therefore we know that:

$$\int_c^d \int_a^b \frac{\partial f}{\partial x} dx dy = \int_c^d f(b, y) - f(a, y) dy = \frac{1}{i} \int_{\gamma_2} f(z) dz + \frac{1}{i} \int_{\gamma_4} f(z) dz$$

Similarly, we can compute:

$$i \int_c^d \int_a^b \frac{\partial f}{\partial y} dx dy$$

Since we are dealing with continuous functions on compact subset, we know that we can either use the Fubini from multivariable calculus or the Fubini from Lebesgue integration and switch the integral order and get:

$$i \int_c^d \int_a^b \frac{\partial f}{\partial y} dy dx = i \int_a^b \int_d^c \frac{\partial f}{\partial y} dy dx$$

Then I will let you do the parametrization and see that this is indeed:

$$\frac{-1}{i} \left( \int_{\gamma_1} f(z) dz + \int_{\gamma_3} f(z) dz \right)$$

□

Using the above lemma and the theorem on bump functions, we get:

**Proposition 1.3.** Let  $R, R'$  be a closed rectangles and let  $R' \subset \mathring{R}$ . If  $f \in C^\infty(R \setminus \mathring{R}')$  again with values in  $\mathbb{C}$  then:

$$\iint_{R \setminus \mathring{R}'} \frac{\partial f}{\partial \bar{z}} dz \wedge d\bar{z} = - \int_{\partial R} f dz + \int_{\partial R'} f dz$$

*Proof.* Clearly if  $f$  is smooth on  $R$ , then we know that we can apply the previous lemma to:

$$\iint_R \frac{\partial f}{\partial \bar{z}} dz \wedge d\bar{z} - \iint_{R'} \frac{\partial f}{\partial \bar{z}} dz \wedge d\bar{z} = - \int_{\partial R} f dz + \int_{\partial R'} f dz$$

However, we only assumed that  $f \in C^\infty(R \setminus \mathring{R}')$ , therefore we need to do some work here.

By our assumption, there is some open  $U \supset R \setminus \mathring{R}'$  such that  $f$  is smooth on  $U$ . Recall that using the theorem about the bump function, there is some function  $\alpha \in C^\infty(U)$  such that  $\alpha = 1$  on  $R \setminus \mathring{R}'$  and  $\alpha$  is supported in  $U$ , then if we consider the function  $g(z)$  which is defined by  $g(z) = \alpha(z) \cdot f(z)$  if  $z \in U$  and is defined to be 0 outside  $U$ . Notice that this definition actually glues to a function  $g$  on all of  $\mathbb{C}$  and you can check that  $g$  is actually in  $C^\infty(\mathbb{C})$ , then we can apply the previous lemma to  $g$  and get:

$$\iint_{R \setminus \mathring{R}'} \frac{\partial g}{\partial \bar{z}} dz \wedge d\bar{z} = \iint_R \frac{\partial g}{\partial \bar{z}} dz \wedge d\bar{z} - \iint_{R'} \frac{\partial g}{\partial \bar{z}} dz \wedge d\bar{z} = - \int_{\partial R} g dz + \int_{\partial R'} g dz$$

However, we know that on  $R \setminus \mathring{R}'$ ,  $g(z) = f(z)$  since  $\alpha = 1$ , then we actually just have:

$$\iint_{R \setminus \mathring{R}'} \frac{\partial f}{\partial \bar{z}} dz \wedge d\bar{z} = - \int_{\partial R} f dz + \int_{\partial R'} f dz$$

□

Here is another little lemma we need:

**Lemma 1.4.**

$$\iint_{|z| < 1} \frac{1}{|z|} dx dy < \infty$$

In other words,  $\frac{1}{|z|} \in L^1(D(0, 1))$ .

*Proof.* Notice that the interval  $[0, 1]$  is just a measure 0 set in  $D(0, 1)$ , we know that the integration can be rewritten as:

$$\iint_{D(0,1) \setminus [0,1]} \frac{1}{|z|} dx dy$$

Then recall that we have the **Lebesgue integration change of variable formula**, where we have a diffeomorphism:

$$(r, \theta) : (0, 1) \times (0, 2\pi) \rightarrow D(0, 1) \setminus [0, 1], \quad (r, \theta) \mapsto r e^{i\theta}$$

Then we have:

$$\iint_{D(0,1) \setminus [0,1]} \frac{1}{|z|} dx dy = \int_0^1 \int_0^{2\pi} d\theta dr = 2\pi$$

□

Then we can state the theorem that gives us the inhomogeneous Cauchy's integral formula we need to prove Runge's theorem:

**Theorem 1.5.** Let  $f \in C_0^\infty(\mathbb{C})$  (a compactly supported smooth function on  $\mathbb{C}$  with values in  $\mathbb{C}$ ), then for any  $\omega \in \mathbb{C}$ , we have:

$$f(\omega) = \frac{1}{2\pi i} \iint_{\mathbb{C}} \frac{\partial f}{\partial \bar{z}} \frac{1}{z - \omega} dz \wedge d\bar{z}$$

Writing it in a different form:

$$-\pi f(\omega) = \iint_{\mathbb{C}} \frac{\partial f}{\partial \bar{z}} \frac{1}{z - \omega} dx dy$$

Notice that this integral actually makes sense since  $\frac{\partial f}{\partial \bar{z}}$  is continuous and compactly supported, thus is integrable on  $\mathbb{C}$ . We also know  $\frac{1}{z - \omega}$  is integrable on  $\mathbb{C}$  by the previous lemma we proved, therefore their product is integrable, or in other words even in  $L^1(\mathbb{C})$ .

*Proof.* Again if we want to prove:

$$-\pi f(\omega) = \iint_{\mathbb{C}} \frac{\partial f}{\partial \bar{z}} \frac{1}{z - \omega} dx dy$$

We now that we can consider the diffeomorphism which translates  $x$  and  $y$  such that after we do this diffeomorphism, we get:

$$\iint_{\mathbb{C}} \frac{\partial f}{\partial \bar{z}}(z + \omega) \frac{1}{z} dx dy$$

Since  $f(z + \omega)$  is still compactly supported, we can choose a closed rectangle  $R$  such that  $f(z + \omega)$  has support contained in  $\mathring{R}$  and  $0 \in R$ .

Then we again choose a  $\epsilon$  small and a rectangle  $R_\epsilon$  with sides length of  $\epsilon$  centered at 0. Then notice that:

$$\iint_{\mathbb{C}} \frac{\partial f}{\partial \bar{z}}(z + \omega) \frac{1}{z} dx dy = \iint_{\mathbb{C}} \frac{\partial}{\partial \bar{z}} \left( \frac{f(z + \omega)}{z} \right) dx dy$$

Indeed, we know that since  $\frac{1}{z}$  is holomorphic on  $\mathbb{C} \setminus \{0\}$ , then  $\frac{\partial}{\partial \bar{z}} \frac{1}{z} = 0$  for  $z \neq 0$ . Then we know that since:

$$\frac{\partial}{\partial \bar{z}} \left( \frac{f(z + \omega)}{z} \right) = \frac{\partial}{\partial \bar{z}} f(z + \omega) + \frac{\partial f}{\partial \bar{z}}(z + \omega) \cdot \frac{1}{z}$$

We know that these two functions agree when  $z \neq 0$ . However we know that  $\{0\}$  is only a measure 0 subset, therefore it doesn't affect the integral. Therefore let's just compute  $\iint_{\mathbb{C}} \frac{\partial}{\partial \bar{z}} \left( \frac{f(z + \omega)}{z} \right) dx dy$ .

Recall that the function:

$$\left| \frac{\partial f(z + \omega)}{\partial \bar{z}} \frac{1}{z} \right| \stackrel{\text{a.e.}}{=} \left| \frac{\partial}{\partial \bar{z}} \left( \frac{f(z + \omega)}{z} \right) \right| \in L^1(\mathbb{C})$$

We can also think about the function:

$$g_\epsilon(z) = \frac{\partial}{\partial \bar{z}} \left( \frac{f(z+\omega)}{z} \right) 1_{R \setminus \dot{R}_\epsilon}$$

where 1 is the indicator function. Then we know that  $g_\epsilon(z)$  converges to  $\frac{f(z+\omega)}{z}$  almost everywhere on  $\mathbb{C}$ , therefore since:

$$|g_\epsilon(z)| \leq \left| \frac{\partial}{\partial \bar{z}} \left( \frac{f(z+\omega)}{z} \right) \right|$$

Using dominated convergence theorem, we know that:

$$\iint_{\mathbb{C}} \frac{\partial}{\partial \bar{z}} \left( \frac{f(z+\omega)}{z} \right) dx dy = \lim_{\epsilon \rightarrow 0} \iint_{\mathbb{C}} g_\epsilon(z) dx dy$$

Then for each  $g_\epsilon(z)$ , we know that the integral is:

$$\iint_{\mathbb{C}} g_\epsilon(z) dx dy = \iint_{R \setminus \dot{R}_\epsilon} \frac{\partial}{\partial \bar{z}} \left( \frac{f(z+\omega)}{z} \right) dx dy = \frac{-1}{2i} \iint_{R \setminus \dot{R}_\epsilon} \frac{\partial}{\partial \bar{z}} \left( \frac{f(z+\omega)}{z} \right)$$

Then using the previous result, we have:

$$\frac{-1}{2i} \iint_{R \setminus \dot{R}_\epsilon} \frac{\partial}{\partial \bar{z}} \left( \frac{f(z+\omega)}{z} \right) dz \wedge d\bar{z} = \frac{-1}{2i} \left( - \int_{\partial R} \left( \frac{f(w+z)}{z} \right) dz + \int_{\partial R_\epsilon} \left( \frac{f(w+z)}{z} \right) dz \right)$$

Notice that by our choice of  $R$ , we know that:

$$- \int_{\partial R} \left( \frac{f(w+z)}{z} \right) dz = 0$$

Therefore the above integral is just:

$$\frac{-1}{2i} \iint_{R \setminus \dot{R}_\epsilon} \frac{\partial}{\partial \bar{z}} \left( \frac{f(z+\omega)}{z} \right) dz \wedge d\bar{z} = \frac{-1}{2i} \int_{\partial R_\epsilon} \frac{f(w+z)}{z} dz$$

However, notice that:

$$\int_{\partial R_\epsilon} \frac{f(w+z)}{z} dz = \int_{\partial R_\epsilon} \frac{f(w+z) - f(w) + f(w)}{z} dz = 2\pi i f(w) + \int_{\partial R_\epsilon} \frac{f(z+w) - f(w)}{z} dz$$

This gives us:

$$\iint_{\mathbb{C}} \frac{\partial}{\partial \bar{z}} \left( \frac{f(z+\omega)}{z} \right) dx dy = \lim_{\epsilon \rightarrow 0} \frac{-1}{2i} \left( \int_{\partial R_\epsilon} \frac{f(z+w) - f(w)}{z} dz + 2\pi i f(w) \right)$$

Notice that since we assumed that  $f$  is smooth on  $\mathbb{C}$ , then we have a bound  $M > 0$  such that:

$$\left| \frac{f(z+w) - f(w)}{z} \right| < M$$

Then we know that we have:

$$\left| \lim_{\epsilon} \int_{\partial R_\epsilon} \frac{f(z+w) - f(w)}{z} dz \right| \leq \lim_{\epsilon} \int_{\partial R_\epsilon} M dz = 0$$

Therefore we eventually get:

$$\iint_{\mathbb{C}} \frac{\partial}{\partial \bar{z}} \left( \frac{f(z+\omega)}{z} \right) dx dy = \pi f(w)$$

which is exactly what we desired. □

This let's us to partially solve the  $\bar{\partial}$  equation:

**Theorem 1.6.** Let  $f \in C_0^\infty(\mathbb{C})$ , then if we define:

$$u(\omega) = \frac{1}{2\pi i} \iint_{\mathbb{C}} \frac{f(z)}{z - \omega} dz \wedge d\bar{z}$$

Then  $u \in C^\infty(\mathbb{C})$  and we have:

$$\frac{\partial u}{\partial \bar{z}}(w) = f(w), \quad \forall w \in \mathbb{C}$$

*Proof.* □

## 2 Runge's Theorem

Let  $\Omega$  be an open subset of  $\mathbb{C}$  and let  $K$  be a compact subset in  $\Omega$ . For any continuous function  $\phi$  on  $K$ , we set:

$$|\phi|_K = \sup_{z \in K} |\phi(z)|$$

Let  $\mathcal{H}(\Omega)$  be the set of all holomorphic functions on  $\Omega$ . We can define a topology on  $\mathcal{H}(\Omega)$  by declaring that a basis of  $f \in \mathcal{H}(\Omega)$  is given by:

$$B_{\epsilon, K} = \{g \in \mathcal{H}(\Omega) : |f - g|_K < \epsilon\}$$

here notice that for each compact subset  $K$  and each  $\epsilon > 0$ , we obtain a  $B_{\epsilon, K}$  and the basis is obtained when  $K$  ranges over all compact subsets and  $\epsilon$  runs over all positive real numbers. This is indeed a basis, since we know that the intersection:

$$B_{\epsilon_1, K_1} \cap B_{\epsilon_2, K_2} = B_{\epsilon_3, K_1 \cup K_2}$$

where  $\epsilon_3$  is the minimum of  $\epsilon_1, \epsilon_2$ . This tells you that the for any finite intersection, there is a smaller element in the system that is contained in the intersection. Then we define the topology on  $\mathcal{H}(\Omega)$  to be those generated by these basis at each point.

This topology is in fact metrizable: Recall that for any open subset of  $\mathbb{C}$ , here we use  $\Omega$ , there exists a sequence of compact subsets  $K_n$  such that  $K_n$  is contained in the interior of  $K_{n+1}$  and  $\cup_n K_n = \Omega$ . Then for  $f, g \in \mathcal{H}(\Omega)$ , we define:

$$d(f, g) = \sum_{n=1}^{\infty} 2^{-n} \frac{|f - g|_{K_n}}{1 + |f - g|_{K_n}}$$

Clearly that by our definition for each  $n$ ,  $\frac{|f - g|_{K_n}}{1 + |f - g|_{K_n}}$ , therefore the above series indeed converges. It's easy to check this is indeed a metric and this metric in fact define the same topology. To see why, we only need to show that at a point, they give equivalent basis. Given a compact set  $K$  and  $\epsilon$ , we know that it's contained in some  $K_N$  in the sequence, then we know that if we consider  $B_{\epsilon, K}$ , we can consider  $\{g : d(f, g) < \epsilon/n\}$ . We know that if  $g$  is in  $\{g : d(f, g) < \epsilon' \cdot 2^{-N}\}$ , then we know that:

$$\sum_n 2^{-n} \frac{|f - g|_{K_n}}{1 + |f - g|_{K_n}} < \epsilon' 2^{-N}$$

In particular, we know that  $\frac{|f - g|_{K_N}}{1 + |f - g|_{K_N}} < \epsilon'$ , this implies:

$$|f - g|_{K_N} < \epsilon' + \epsilon' |f - g|_{K_N} \Rightarrow |f - g|_{K_N} < \frac{\epsilon'}{1 - \epsilon'}$$

Then we know that if we take  $\epsilon'$  to be small enough, we have  $\frac{\epsilon'}{1 - \epsilon'} < \epsilon$ . Conversely, if we suppose that:

$$\sum_n 2^{-n} \frac{|f - g|_{K_n}}{1 + |f - g|_{K_n}} < \epsilon$$

Then we know that there is at least one  $N$  such that  $|f - g|_{K_N} < \epsilon$ , then we know that we can just take  $B_{K_N, \epsilon}$ .

This metric also makes  $\mathcal{H}(\Omega)$  complete. Indeed, if we assume that  $\{f_n\}$  is a Cauchy sequence, which means that for any  $2^{-2M}$  there is  $N$  such that all  $m, n > N$  we have:

$$d(f_n - f_m) = \sum_{i=1}^{\infty} 2^{-i} \frac{|f_n - f_m|_{K_i}}{1 + |f_n - f_m|_{K_i}} < 2^{-M}$$

Then this implies that:

$$\sum_i 2^{M-i} \frac{|f_n - f_m|_{K_i}}{1 + |f_n - f_m|_{K_i}} < 2^{-M}, \quad \forall m, n$$

This implies that we must have  $\frac{|f_n - f_m|_{K_M}}{1 + |f_n - f_m|_{K_M}} < 2^{-M}$  which roughly means that  $|f_n - f_m|_{K_M} < 2^{-M}$ . Now if  $K$  is a compact subset of  $\Omega$ , we know that when we take  $M$  to be large enough  $K \subset K_M \subset K_{M+1} \subset \dots$  and we have:

$$|f_m - f_n|_K < 2^{-M}$$

Therefore we know that for any  $\epsilon > 0$ , there is a  $N$  such that for all  $m, n > N$ , we have:

$$|f_m - f_n|_K < \epsilon$$

Therefore we know that  $f_n$  converges uniformly on each compact subset of  $\Omega$ , which means that the limit function is also holomorphic. Therefore we know that this is indeed a complete metric space.

The above topology on  $\mathcal{H}(\Omega)$  is often called the topology of uniform convergence on compact sets. It's called this name because if we consider a sequence of holomorphic functions  $\{f_n\}$ , then it converges to an element in  $\mathcal{H}(\Omega)$  iff this sequence converges uniformly on any compact subset of  $\Omega$ . I will let you to fill in the details, and it shouldn't be too hard given our above discussion.

Now let  $K$  be a compact subset of  $\mathbb{C}$ , we use  $\mathcal{O}(K)$  to denote the space of continuous functions  $\phi$  on  $K$  such that there is an open subset  $U$  of  $\mathbb{C}$  which contains  $K$  and a function  $f \in \mathcal{H}(U)$  with  $f|_K = \phi$ . That is to say that  $f$  is the restriction of some holomorphic function on a larger open subset.

Finally, we will use  $C(K)$  to denote the space of all continuous functions on  $K$  with the norm  $\|\phi\| = |\phi|_K$ . Then this makes  $C(K)$  a Banach-space over  $\mathbb{C}$  (since uniform limits of continuous functions on  $K$  is still continuous). We will consider  $\mathcal{O}(K)$  as a subspace of  $C(K)$  with the same norm. However, it's usually not closed.

Finally, if  $\Omega$  is open in  $\mathbb{C}$  and  $K$  is compact in  $\Omega$ , we denote by  $\rho = \rho_{\Omega, K}$  the restriction map:

$$\rho : \mathcal{H}(\Omega) \rightarrow \mathcal{O}(K), \quad \rho(f) = f|_K$$

Then we have the first form of Runge's Theorem:

**Theorem 2.1. (Runge's Theorem: First Form)** Let  $\Omega$  be open in  $\mathbb{C}$  and let  $K$  be a compact subset of  $\Omega$ , then the following statements are equivalent:

1.  $\rho(\mathcal{H}(\Omega))$  is dense in  $\mathcal{O}(K)$ .
2. No connected components of  $\Omega \setminus K$  is relatively compact in  $\Omega$ .
3. For any  $a \in \Omega$  and  $a \notin K$ , there exists a  $f \in \mathcal{H}(\Omega)$  with:

$$|f(a)| > |f|_K$$

*Proof.* Let first deal with  $1 \Rightarrow 2$ : Recall that a set in  $\Omega$  is relatively compact if it's closure is compact. Suppose that  $\Omega \setminus K$  has a component  $U$  which is relatively compact. let's prove that  $\partial U \subset K$ . To see why, let  $a \in \partial U$  and for the sake of contradiction, we assume  $a \notin K$ . Since it's not in  $K$ , there is a disk  $D$  around of  $a$  such that  $D \subset \Omega \setminus K$  and since  $a$  is in the boundary of  $U$ , we know that there is a point in  $U$  such that it's also in the disk. Consequently,  $D \cup U$  is connected, but  $U$  is already the connected component, therefore  $U \subset U$ , which means that  $D \cap K = \emptyset$ , therefore  $a \notin \partial U$ .

Now let  $z_0 \in U$  and let  $f(z) = \frac{1}{(z-z_0)}$  which is holomorphic on  $\Omega \setminus \{z_0\} \supset K$ , which implies that  $f|_K \in \mathcal{O}(K)$ , then suppose that 1 is true for the sake of contradiction, then there will be a sequence of function  $\{f_n\}$  in  $\mathcal{H}(\Omega)$  such that after you restrict them to  $K$  they converges uniformly to  $f|_K$ . However, by the maximum principle, we know:

$$\sup_{\overline{U}} |f_n - f_m| = \sup_{\partial U} |f_n - f_m| \leq \sup_K |f_n - f_m|$$

However, we already know that on  $K$ ,  $f_n$  uniformly converges to  $f|_K$ , therefore  $\sup_K |f_n - f_m|$  goes to 0 if  $m, n$  goes to infinity, therefore we know that  $f_n$  also converges uniformly on  $\overline{U}$ , to a holomorphic function  $g \in \mathcal{H}(U)$ . Note that we are using the assumption that  $U$  is relatively compact to apply the maximum principle which is only valid if the open set is relatively compact.

Now think about  $(z - z_0)f_n$ , which converges to 1 uniformly on  $\partial U \subset K$ , then again by the maximum principle we know that  $(z - z_0)f_n$  converges to 1 uniformly on  $U$ . This implies that:

$$(z - z_0)g(z) = 1, \forall z \in U$$

But this absurd since we know for  $z = z_0$ , the function is 0. This contradiction tells you  $1 \Rightarrow 2$ .

Then we deal with  $2 \Rightarrow 1$ . Let  $\rho(\mathcal{H}(\Omega))$  and we know  $E \subset \mathcal{O}(K) \subset C(K)$ . Then by the Hahn-Banach theorem applied to  $C(K)$ , we know that  $E$  is dense in  $\mathcal{O}(K)$  iff the following holds:

Let  $\lambda$  be a continuous linear functional on  $C(K)$  then if  $\lambda|_E = 0$ , we have  $\lambda|_{\mathcal{O}(K)} = 0$ . For those that are not too familiar with functional analysis, we state the version of Hahn-Banach we are using here.

**Theorem 2.2. (Hahn-Banach Theorem)** Suppose that  $X$  is a normed vector space (not necessarily complete) over  $\mathbb{C}$  with  $Y$  a subspace. If  $f : Y \rightarrow \mathbb{C}$  is a continuous linear functional then  $f$  extends to a continuous functional on  $X$ .

One consequence of this theorem is what we are using above in the proof:

**Corollary 2.3.** Let  $M$  be a subspace of a normed vector space  $X$  over  $\mathbb{C}$  and let  $x_0$  be a point in  $X$  not in  $\overline{M}$ . Then there is a continuous linear functional  $\Lambda \in X^*$  such that:

$$\Lambda|_M = 0, \quad \Lambda(x_0) = 1$$

The way you prove this corollary using Hahn-Banach is to define a linear functional  $f : \overline{M} + \mathbb{C}x_0$  by  $f(m + ax_0) = a$ , then you check that this is a continuous functional on  $\overline{M} + \mathbb{C}x_0$  then use Hahn-Banach to extend to  $X$ . Then we go back to the proof:

This corollary above says that if every continuous linear function  $\lambda$  defined on  $C(K)$  which is zero on  $E$  is also 0 on  $\mathcal{O}(K)$ , then if  $\overline{E}$  is not  $\mathcal{O}(K)$ , then we can create a counter-example using the above corollary, which is a contradiction.

Now suppose that  $\lambda$  is a continuous linear functional on  $C(K)$ , we can define a function:

$$\Phi_\lambda(w) = \lambda(\phi_w)$$

where  $\phi_w = \frac{1}{z-\omega}$  and  $z \in K$ . I claim that  $\Phi_\lambda$  is in fact in  $\mathcal{H}(\mathbb{C} \setminus K)$  and  $\Phi_\lambda^{(n)}(w) = n! \lambda(\phi_{w,n})$  where  $\phi_{w,n} = \frac{1}{(z-w)^{n+1}}$ . To see why, for any  $a \notin K$ , we can choose  $r > 0$  such that  $\overline{D(a, r)} \cap K = \emptyset$ , therefore we know that:

$$\phi_w(z) = \frac{1}{z-\omega} = \frac{1}{z-a} \frac{1}{1 - \frac{w-a}{z-a}}$$

We know that for  $|\omega - a| < r$  and  $z \in K$ , the series:

$$\sum_{n=0}^{\infty} \frac{(w - a)^n}{(z - a)^{n+1}}$$

converges uniformly for all  $z \in K$ , then we know that the following series converges uniformly to  $\phi_w(z)$  for  $|w - a| > r$ :

$$\phi_w(z) = \sum_n \frac{(w - a)^n}{(z - a)^{n+1}}$$

Then we know that since  $\lambda$  is continuous and  $\frac{1}{(z-a)^{n+1}}$  is in  $C(K)$  since  $a \notin K$ , then we know:

$$\lambda(\phi_w(z)) = \sum_n (w - a)^n \lambda((z - a)^{-n-1})$$

Therefore we know that at least in  $|w - a| < r$ , we have a power series representation:

$$\Phi_\lambda(w) = \sum_n (w - a)^n \lambda(\phi_{a,n})$$

Then we know that  $\Phi_\lambda \in D(a, r)$  with  $\Phi^{(n)}(a) = n! \lambda(\phi_{a,n})$ . Since this holds for each  $a \notin K$ , we know that  $\Phi_\lambda \in \mathcal{H}(\mathbb{C} \setminus K)$ .

Suppose that  $\lambda$  is a linear functional on  $C(E)$  with  $\lambda|_E = 0$ . I claim this implies that  $\Phi_\lambda = 0$  on  $\mathbb{C} \setminus K$  if  $\Omega \setminus K$  has no connected component that is relatively compact in  $\Omega$ . To show this,  $\square$

## References

- [NN01] Raghavan Narasimhan and Yves Nievergelt, *Complex analysis in one variable*, 2 ed., Birkhäuser, Boston, 2001.